

New Wheels Project

Introduction to SQL

BRIANA SALINAS

2/6/2026

Problem Statement

Business Context

A lot of people in the world share a common desire: to own a vehicle. A car or an automobile is seen as an object that gives the freedom of mobility. Many now prefer pre-owned vehicles because they come at an affordable cost, but at the same time, they are also concerned about whether the after-sales service provided by the resale vendors is as good as the care you may get from the actual manufacturers.

New-Wheels, a vehicle resale company, has launched an app with an end-to-end service from listing the vehicle on the platform to shipping it to the customer's location. This app also captures the overall after-sales feedback given by the customer.

Objective

New-Wheels sales have been dipping steadily in the past year, and due to the critical customer feedback and ratings online, there has been a drop in new customers every quarter, which is concerning to the business. The CEO of the company now wants a quarterly report with all the key metrics sent to him so he can assess the health of the business and make the necessary decisions.

As a data analyst, you see that there is an array of questions that are being asked at the leadership level that need to be answered using data. Import the dump file that contains various tables that are present in the database. Use the data to answer the questions posed and create a quarterly business report for the CEO.

Question 1: Find the total number of customers who have placed orders. What is the distribution of the customers across states?

Solution Query:

```
-- 1.a
SELECT
    COUNT(DISTINCT c.customer_id) AS total_customers
FROM customer_t AS c
JOIN order_t AS o ON c.customer_id = o.customer_id;

-- 1.b
SELECT state, COUNT(customer_id) AS total_customers
FROM customer_t
GROUP BY state
ORDER BY total_customers desc;
```

Output:

Test Cases

Run SQL

Result: Passed

Query 1

Query:

SELECT
COUNT(DISTINCT c.customer_id) AS total_customers
FROM customer_t AS c
JOIN order_t AS o
ON c.customer_id = o.customer_id

Output:

Showing 1 rows

total_customers
994

Test Cases
Run SQL

Result: Passed

Query 1

Query 2

Query:

SELECT state, COUNT(customer_id) AS total_customers
FROM customer_t
GROUP BY state
ORDER BY total_customers desc

Output:

Showing first 10 rows out of 49 rows

state	total_customers
Texas	97
California	97
Florida	86
New York	69
District of Columbia	35
Ohio	33
Colorado	33
Alabama	29
Washington	28
Arizona	26

Observations and Insights:

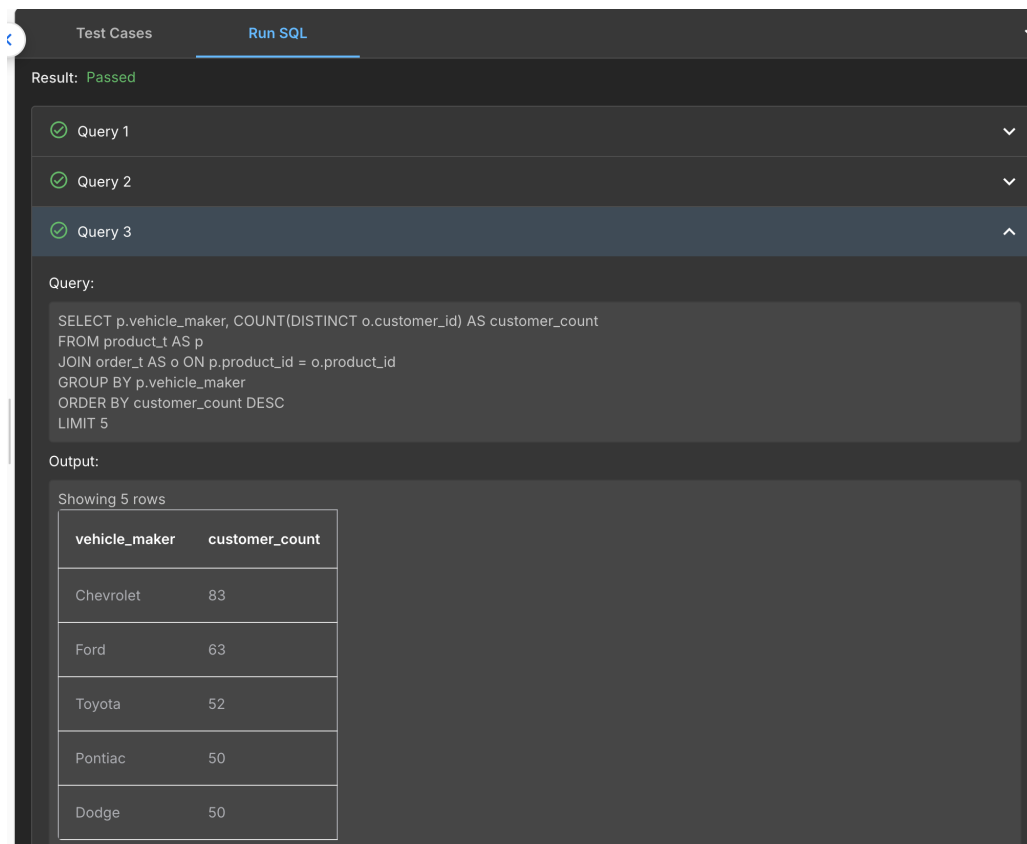
- Texas and California each have 97 customers, the highest in the dataset, followed closely by Florida and New York. This indicates that customer purchases are strongest in high population, economically active states.
- States with smaller customer counts, such as Arizona, Washington, and Alabama, represent areas where the company could potentially increase customer acquisition through targeted promotions or advertising.
- There are 994 customers who have placed orders, which indicates that all customers in the dataset are active purchasers. This suggests strong engagement and no inactive customer records within the system. Whether I chose DISTINCT or without it in the code, customers remained 994.

Question 2: Which are the top 5 vehicle makers preferred by the customers?

Solution Query:

```
SELECT p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count
FROM product_t AS p
JOIN order_t AS o ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5;
```

Output:



The screenshot shows a SQL query execution interface. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the result status is 'Passed'. A list of queries is shown, with 'Query 3' selected. The query text is displayed in a text area, and the output is shown in a table below it.

Result: Passed

Query: 1, 2, 3 (Query 3 is selected)

Query:

```
SELECT p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count
FROM product_t AS p
JOIN order_t AS o ON p.product_id = o.product_id
GROUP BY p.vehicle_maker
ORDER BY customer_count DESC
LIMIT 5
```

Output:

Showing 5 rows

vehicle_maker	customer_count
Chevrolet	83
Ford	63
Toyota	52
Pontiac	50
Dodge	50

Observations and Insights:

- Chevrolet has the highest number of distinct customers (83) making it the most preferred vehicle maker in the dataset. This indicates strong customer demand and brand popularity.
- Pontiac and Dodge both have 50 customers, indicating equal customer preference at that rank. This suggests similar brand appeal or availability among customers.
- With 63 customers, Ford ranks second, showing solid market presence. While not leading, it maintains a meaningful gap above most other makers.

Question 3: Which is the most preferred vehicle maker in each state?

Solution Query:

```
SELECT state, vehicle_maker, customer_count
FROM (SELECT c.state, p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count,
      RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT o.customer_id) DESC) AS rnk
FROM product_t AS p
JOIN order_t AS o ON p.product_id = o.product_id
JOIN customer_t AS c ON o.customer_id = c.customer_id
GROUP BY c.state, p.vehicle_maker) AS ranked
WHERE rnk = 1
ORDER BY state;
```

Output:

Test Cases
Run SQL

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query:

```
SELECT state, vehicle_maker, customer_count
FROM (SELECT c.state, p.vehicle_maker, COUNT(DISTINCT o.customer_id) AS customer_count,
      RANK() OVER (PARTITION BY c.state ORDER BY COUNT(DISTINCT o.customer_id) DESC) AS rnk
FROM product_t AS p
JOIN order_t AS o ON p.product_id = o.product_id
JOIN customer_t AS c ON o.customer_id = c.customer_id
GROUP BY c.state, p.vehicle_maker) AS ranked
WHERE rnk = 1
ORDER BY state
```

Output:

Showing first 10 rows out of 143 rows

state	vehicle_maker	customer_count
Alabama	Dodge	5
Alaska	Chevrolet	2
Arizona	Pontiac	3
Arizona	Cadillac	3
Arkansas	Volkswagen	1
Arkansas	Suzuki	1
Arkansas	Pontiac	1
Arkansas	Mitsubishi	1
Arkansas	GMC	1
Arkansas	Chevrolet	1

Observations and Insights:

- The results show that the most preferred vehicle maker is not the same nationwide. Different brands lead in different states, indicating that customer preferences are influenced by regional factors.
- Arkansas is spread equally about vehicle makers tying with 1 customer for 6 vehicles. This suggests evenly split brand preference in certain markets rather than a single dominant maker.
- States such as Alaska and Arkansas have very small top customer counts (like 1–2 customers), indicating lower demand or a smaller customer base in those regions.
- Since vehicle preference differs by state, the company could tailor promotions, inventory stocking, or dealership partnerships based on the most preferred brand in each region.

Question 4: Find the overall average rating given by the customers.

What is the average rating in each quarter?

Consider the following mapping for ratings: “Very Bad”: 1, “Bad”: 2, “Okay”: 3, “Good”: 4, “Very Good”: 5

Solution Query:

```
-- 4.a
SELECT
    AVG(rating_score) AS overall_avg_rating
FROM (SELECT
    CASE customer_feedback
        WHEN 'Very Bad' THEN 1
        WHEN 'Bad' THEN 2
        WHEN 'Okay' THEN 3
        WHEN 'Good' THEN 4
        WHEN 'Very Good' THEN 5
        ELSE NULL
    END AS rating_score
FROM order_t) AS ratings;

-- 4.b
SELECT quarter_number, ROUND(AVG(rating_score),3) AS avg_rating
FROM ( SELECT quarter_number,
    CASE customer_feedback
        WHEN 'Very Bad' THEN 1
        WHEN 'Bad' THEN 2
        WHEN 'Okay' THEN 3
        WHEN 'Good' THEN 4
        WHEN 'Very Good' THEN 5
        ELSE NULL
    END AS rating_score
FROM order_t) AS ratings
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Test Cases Run SQL

Result: Passed

- Query 1
- Query 2
- Query 3
- Query 4
- Query 5

Query:

```
SELECT
  AVG(rating_score) AS overall_avg_rating
FROM (SELECT
  CASE customer_feedback
    WHEN 'Very Bad' THEN 1
    WHEN 'Bad' THEN 2
    WHEN 'Okay' THEN 3
    WHEN 'Good' THEN 4
    WHEN 'Very Good' THEN 5
    ELSE NULL
  END AS rating_score
FROM order_t) AS ratings
```

Output:

Showing 1 rows

overall_avg_rating
3.135

Test Cases Run SQL

Result: Passed

- Query 1
- Query 2
- Query 3
- Query 4
- Query 5
- Query 6

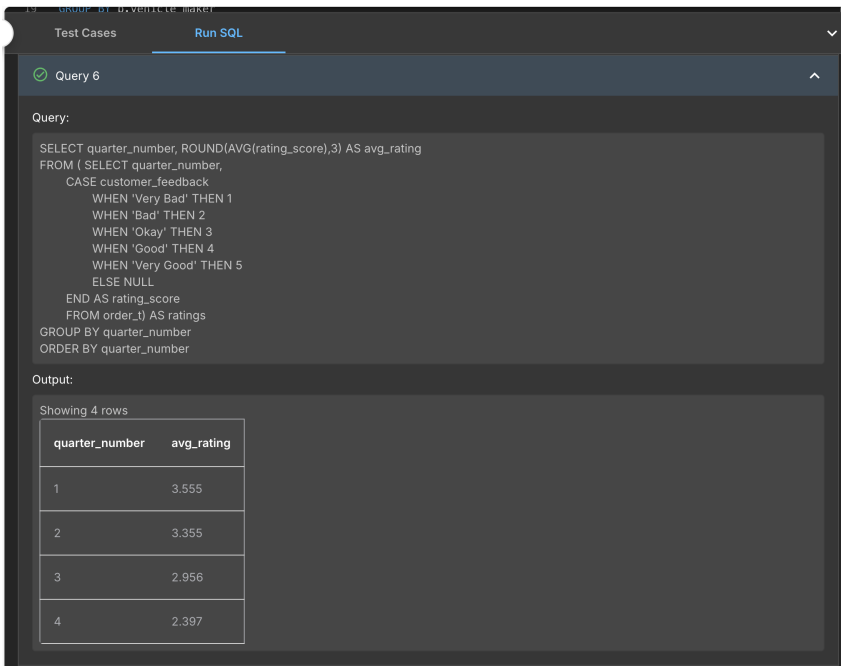
Query:

```
SELECT quarter_number, ROUND(AVG(rating_score),3) AS avg_rating
FROM ( SELECT quarter_number,
  CASE customer_feedback
    WHEN 'Very Bad' THEN 1
    WHEN 'Bad' THEN 2
    WHEN 'Okay' THEN 3
    WHEN 'Good' THEN 4
    WHEN 'Very Good' THEN 5
    ELSE NULL
  END AS rating_score
FROM order_t) AS ratings
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_rating
1	3.135
2	3.135
3	3.135
4	3.135



The screenshot shows a SQL query execution window. The query is labeled 'Query 6' and is a SELECT statement that calculates the average rating for each quarter, rounded to three decimal places. The output shows four rows of data, indicating a consistent downward trend in customer satisfaction from Q1 to Q4.

Query:

```
SELECT quarter_number, ROUND(AVG(rating_score),3) AS avg_rating
FROM ( SELECT quarter_number,
CASE customer_feedback
  WHEN 'Very Bad' THEN 1
  WHEN 'Bad' THEN 2
  WHEN 'Okay' THEN 3
  WHEN 'Good' THEN 4
  WHEN 'Very Good' THEN 5
  ELSE NULL
END AS rating_score
FROM order_t) AS ratings
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	avg_rating
1	3.555
2	3.355
3	2.956
4	2.397

Observations and Insights:

- The overall average rating of 3.135 indicates that customer satisfaction is slightly above neutral ('Okay') but not strongly positive.
- Average ratings decreased each quarter from 3.555 in Q1 to 2.397 in Q4, showing a consistent downward trend in customer satisfaction.
- While the rating declines gradually from Q1 to Q3, the drop into Q4 is more significant (by 1.153), suggesting customer experience may have worsened more sharply toward the end of the year.
- The steady decrease in ratings could reflect issues such as product quality, delivery experience, pricing, or customer service, and may require further investigation that we could find out in the next questions.

Question 5: Find the percentage distribution of feedback from the customers. Are customers getting more dissatisfied over time?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*), 2)
AS very_bad_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*), 2) AS
bad_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*), 2) AS
okay_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*), 2) AS
good_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*),
2) AS very_good_pct
FROM (SELECT quarter_number, customer_feedback
      FROM order_t) AS rating_pct
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Test Cases
Run SQL

Result: Passed

Query 1

Query 2

Query 3

Query 4

Query 5

Query 6

Query 7

Query:

```
SELECT
    quarter_number,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*), 2) AS very_bad_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*), 2) AS bad_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*), 2) AS okay_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*), 2) AS good_pct,
    ROUND(COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*), 2) AS very_good_pct
FROM (SELECT quarter_number, customer_feedback
      FROM order_t) AS rating_pct
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Test Cases
Run SQL

Query 7

Query:

```

SELECT
  quarter_number,
  ROUND(COUNT(CASE WHEN customer_feedback = 'Very Bad' THEN 1 END) * 100.0 / COUNT(*), 2) AS very_bad_pct,
  ROUND(COUNT(CASE WHEN customer_feedback = 'Bad' THEN 1 END) * 100.0 / COUNT(*), 2) AS bad_pct,
  ROUND(COUNT(CASE WHEN customer_feedback = 'Okay' THEN 1 END) * 100.0 / COUNT(*), 2) AS okay_pct,
  ROUND(COUNT(CASE WHEN customer_feedback = 'Good' THEN 1 END) * 100.0 / COUNT(*), 2) AS good_pct,
  ROUND(COUNT(CASE WHEN customer_feedback = 'Very Good' THEN 1 END) * 100.0 / COUNT(*), 2) AS very_good_pct
FROM (SELECT quarter_number, customer_feedback
      FROM order_t) AS rating_pct
GROUP BY quarter_number
ORDER BY quarter_number

```

Output:

Showing 4 rows

quarter_number	very_bad_pct	bad_pct	okay_pct	good_pct	very_good_pct
1	10.97	11.29	19.03	28.71	30
2	14.89	14.12	20.23	22.14	28.63
3	17.9	22.71	21.83	20.96	16.59
4	30.65	29.15	20.1	10.05	10.05

Observations and Insights:

- The combined percentage of negative feedback (“Very Bad” + “Bad”) rises steadily from Q1 to Q4. This confirms that customers are becoming more dissatisfied as the year progresses. In Q1, positive feedback (“Good” + “Very Good”) dominates the distribution. By Q4, negative feedback becomes the majority, showing a clear shift in customer sentiment.
- “Very Good” ratings drop from 30.00% in Q1 to 10.05% in Q4, indicating a sharp reduction in highly satisfied customers.
- The percentage of “Okay” feedback stays around ~19–22% across all quarters. This suggests that neutral customer experiences remain consistent, while extreme positive and negative views shift more dramatically.

Question 6: What is the trend of the number of orders by quarter?

Solution Query:

```
SELECT quarter_number,COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:

Test Cases Run SQL

Result: Passed

- Query 1
- Query 2
- Query 3
- Query 4
- Query 5
- Query 6
- Query 7
- Query 8

Query:

```
SELECT quarter_number,COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Test Cases Run SQL

- Query 7
- Query 8

Query:

```
SELECT quarter_number,COUNT(order_id) AS total_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	total_orders
1	310
2	262
3	229
4	199

Observations and Insights:

- The number of orders decreases every quarter, dropping from 310 in Q1 to 199 in Q4, indicating a steady downward trend in sales activity.
- There is a decline of 111 orders from Q1 to Q4, representing a substantial reduction in customer purchasing volume.
- The consistent quarter over quarter drop suggests that customer demand is softening rather than fluctuating seasonally.
- This downward order trend mirrors the earlier decline in customer satisfaction ratings, suggesting that worsening customer experience may be influencing purchasing behavior.

Question 7: Calculate the net revenue generated by the company.

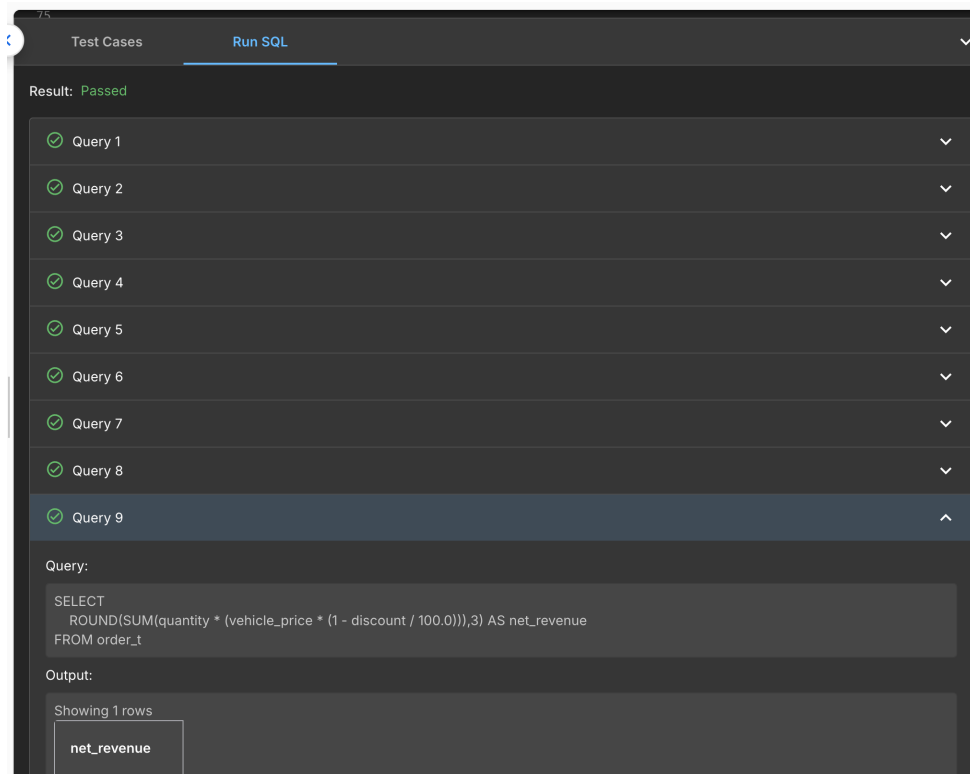
What is the quarter-over-quarter % change in net revenue?

Solution Query:

```
-- 7.a
SELECT
    ROUND(SUM(quantity * (vehicle_price * (1 - discount / 100.0))),3) AS net_revenue
FROM order_t;

-- 7.b
SELECT quarter_number, net_revenue,
    ROUND((net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
        * 100.0
        / LAG(net_revenue) OVER (ORDER BY quarter_number),2) AS percent_diff
FROM (SELECT quarter_number,
    ROUND(SUM(quantity * (vehicle_price * (1 - discount / 100.0))),3) AS net_revenue
    FROM order_t
    GROUP BY quarter_number) revenue_by_quarter
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL execution environment with a 'Test Cases' tab and a 'Run SQL' button. Below the button, it says 'Result: Passed'. A list of 9 queries is shown, all with green checkmarks indicating they passed. Query 9 is selected, and its SQL code is displayed: `SELECT ROUND(SUM(quantity * (vehicle_price * (1 - discount / 100.0))),3) AS net_revenue FROM order_t`. Below the query, it says 'Output: Showing 1 rows' and a table with one column 'net_revenue' is visible.

Showing 1 rows

net_revenue
124714086.324

Query 10

Test Cases Run SQL

Query 9

Query 10

Query:

```
SELECT quarter_number, net_revenue,
ROUND((net_revenue - LAG(net_revenue) OVER (ORDER BY quarter_number))
* 100.0
/ LAG(net_revenue) OVER (ORDER BY quarter_number),2) AS percent_diff
FROM (SELECT quarter_number,
ROUND(SUM(quantity * (vehicle_price * (1 - discount / 100.0))),3) AS net_revenue
FROM order_t
GROUP BY quarter_number) revenue_by_quarter
ORDER BY quarter_number
```

Output:

Showing 4 rows

quarter_number	net_revenue	percent_diff
1	39421580.159	
2	32715830.34	-17.01
3	29229896.194	-10.66
4	23346779.631	-20.13

Observations and Insights:

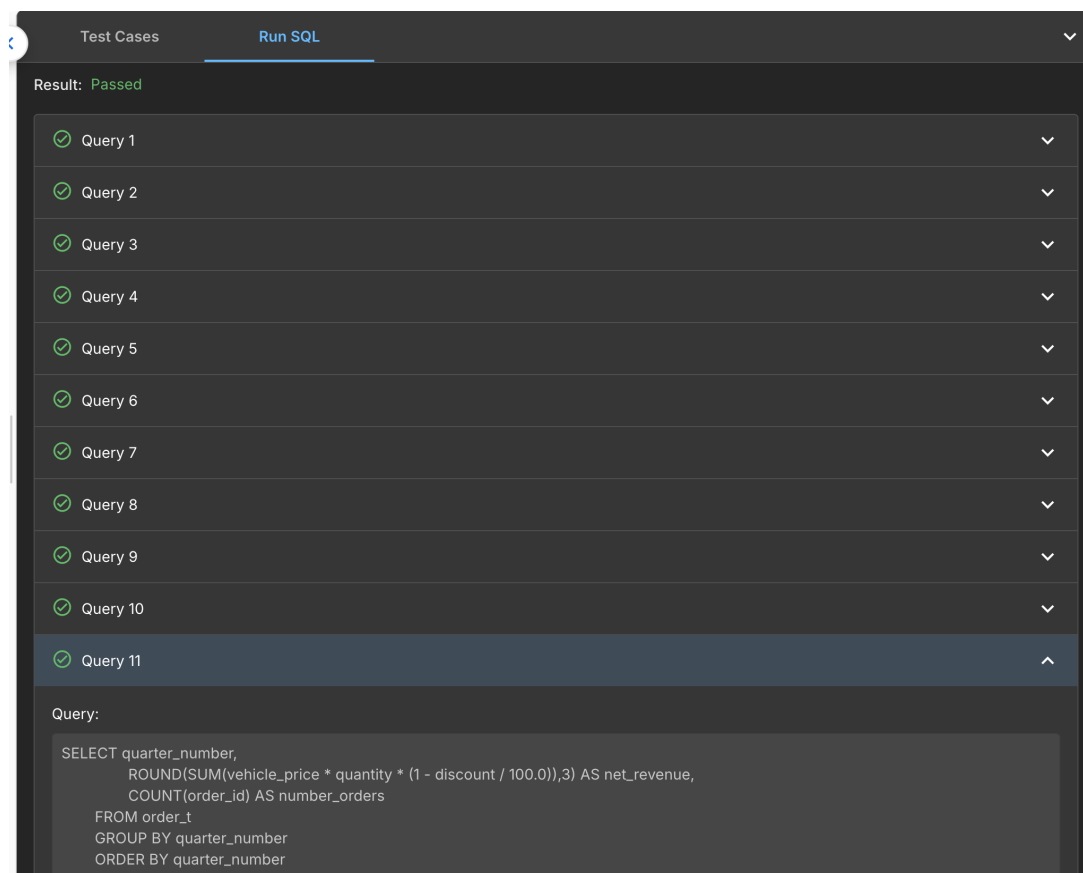
- Total net revenue of approximately \$124.7M indicates that the business operates at a large sales scale despite recent declines.
- Net revenue decreases consistently from \$39.4M in Q1 to \$23.3M in Q4, showing a clear downward financial trend.
- The steepest quarter over quarter decline happens between Q3 and Q4 at -20.13%, indicating accelerated revenue loss late in the year.
- Earlier analysis showed that total orders also decreased each quarter. This suggests that revenue decline is primarily driven by reduced purchasing activity rather than pricing or discount changes alone.

Question 8: What is the trend of net revenue and orders by quarters?

Solution Query:

```
SELECT quarter_number,  
       ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)),3) AS net_revenue,  
       COUNT(order_id) AS number_orders  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number;
```

Output:



Test Cases Run SQL

Result: Passed

- Query 1
- Query 2
- Query 3
- Query 4
- Query 5
- Query 6
- Query 7
- Query 8
- Query 9
- Query 10
- Query 11

Query:

```
SELECT quarter_number,  
       ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)),3) AS net_revenue,  
       COUNT(order_id) AS number_orders  
FROM order_t  
GROUP BY quarter_number  
ORDER BY quarter_number
```

Test Cases
Run SQL

Query 9
Query 10
Query 11

Query:

```

SELECT quarter_number,
       ROUND(SUM(vehicle_price * quantity * (1 - discount / 100.0)),3) AS net_revenue,
       COUNT(order_id) AS number_orders
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number

```

Output:

Showing 4 rows

quarter_number	net_revenue	number_orders
1	39421580.159	310
2	32715830.34	262
3	29229896.194	229
4	23346779.631	199

Observations and Insights:

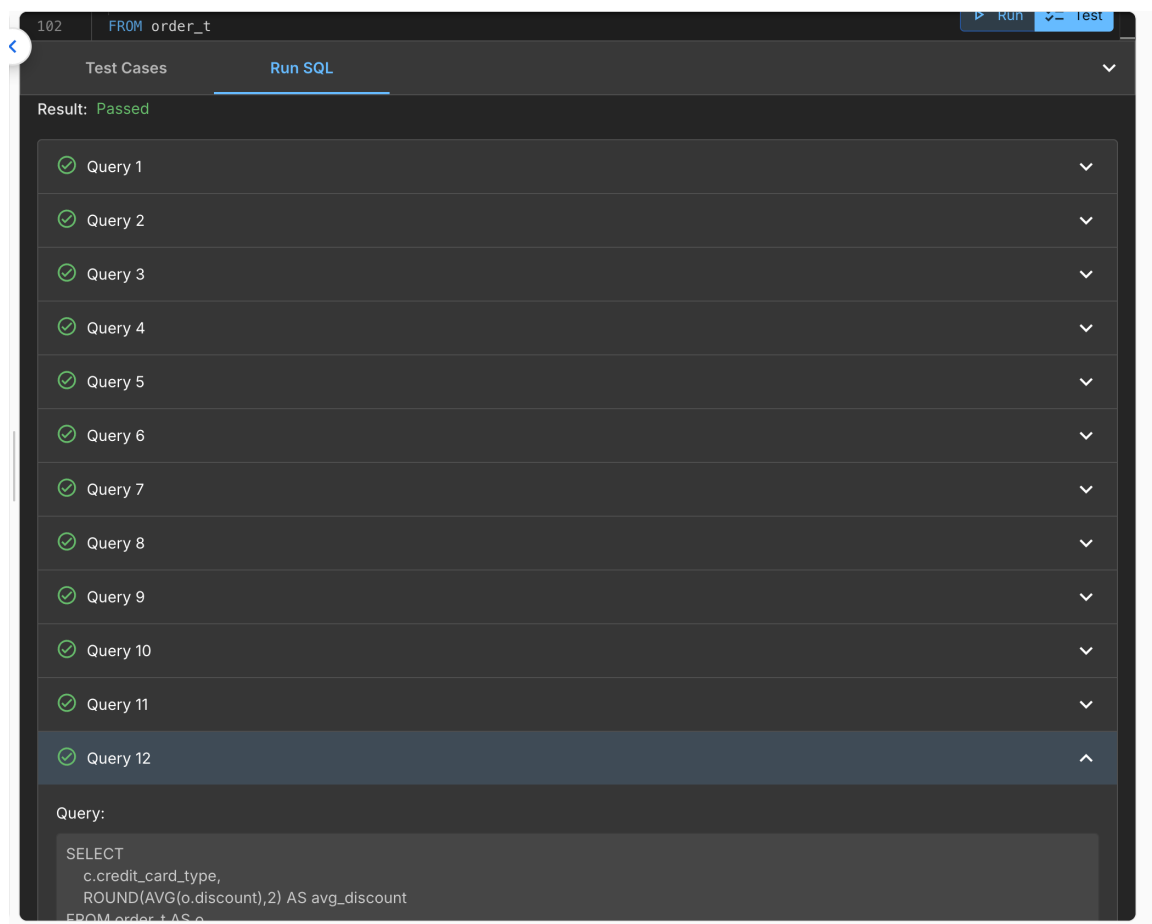
- Net revenue decreases each quarter in line with the decrease in total orders, indicating a direct correlation/relationship between purchasing activity and revenue performance.
- From Q1 to Q2, orders decrease by 48, and net revenue declines by approximately \$6.7M, making it the most significant early drop.
- Order reductions shrink from:
 - Q1 to Q2 = 48
 - Q2 to Q3 = 33
 - Q3 to Q4 = 30
- Those reductions are showing that while demand continues to fall, the rate of decline becomes slightly less steep over time.
- The downward trend in both orders and revenue aligns with earlier findings around declining customer ratings and increasing dissatisfaction, reinforcing concerns about overall business health.

Question 9: What is the average discount offered for different types of credit cards?

Solution Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount),2) AS avg_discount
FROM order_t AS o
JOIN customer_t AS c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY c.credit_card_type;
```

Output:



102 FROM order_t

Test Cases Run SQL

Result: Passed

- Query 1
- Query 2
- Query 3
- Query 4
- Query 5
- Query 6
- Query 7
- Query 8
- Query 9
- Query 10
- Query 11
- Query 12

Query:

```
SELECT
    c.credit_card_type,
    ROUND(AVG(o.discount),2) AS avg_discount
FROM order_t AS o
```

102 FROM order_t

Test Cases Run SQL

```
FROM order_t AS o
JOIN customer_t AS c ON o.customer_id = c.customer_id
GROUP BY c.credit_card_type
ORDER BY c.credit_card_type
```

Output:

Showing first 10 rows out of 16 rows

credit_card_type	avg_discount
americanexpress	0.62
bankcard	0.61
china-unionpay	0.62
diners-club-carte-blanc	0.61
diners-club-enroute	0.6
diners-club-international	0.58
diners-club-us-ca	0.61
instapayment	0.62
jcb	0.61
laser	0.64

Observations and Insights:

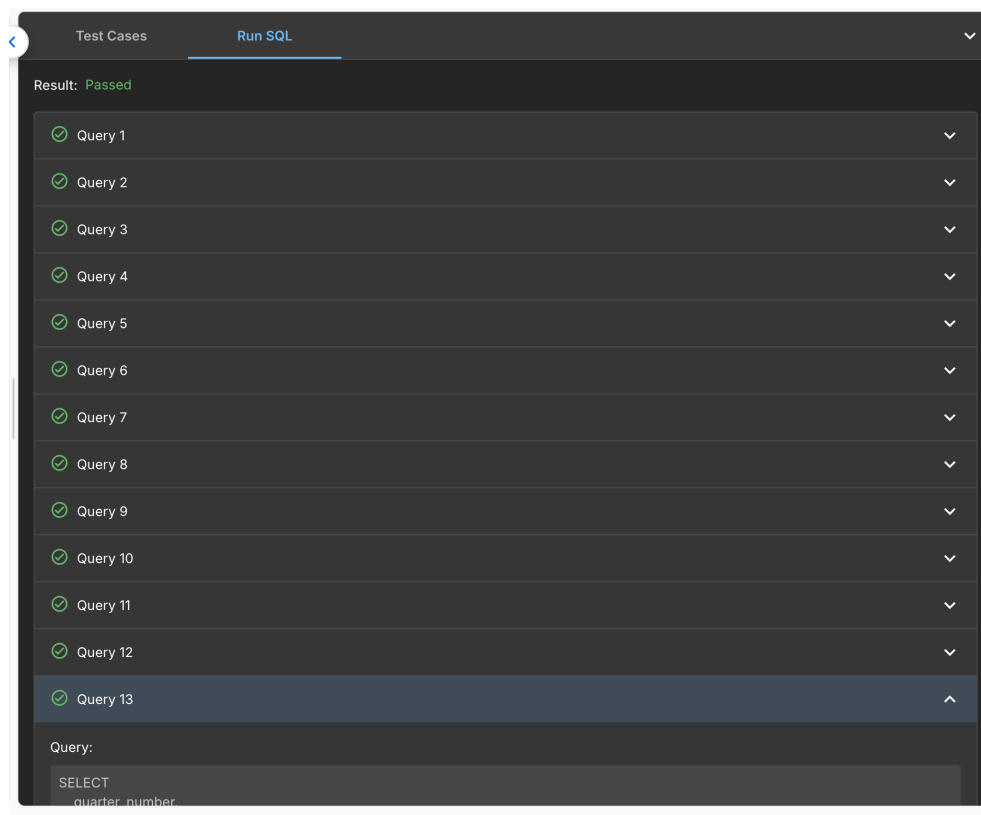
- Most credit card types receive average discounts between 0.60% and 0.62%, indicating a standardized discount structure rather than highly different offers.
- The difference between the lowest in the first 10 rows of data (Diners Club International ~0.58%) and highest (Laser ~0.64%) average discount is minimal, suggesting limited pricing incentive differences across payment methods.
- Because discount differences are small, customers may not be strongly influenced to choose one credit card over another based on promotional savings alone.

Question 10: What is the average time taken to ship the placed orders for each quarter?

Solution Query:

```
SELECT
    quarter_number,
    ROUND(AVG(julianday(ship_date) - julianday(order_date)),2) AS avg_days_to_ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number;
```

Output:



The screenshot shows a SQL execution interface with a dark theme. At the top, there are tabs for 'Test Cases' and 'Run SQL'. Below the tabs, the status 'Result: Passed' is displayed. A list of 13 test cases, labeled 'Query 1' through 'Query 13', is shown, each with a green checkmark icon and a dropdown arrow. 'Query 13' is currently selected and highlighted. Below the list, there is a 'Query:' label and a text area containing the SQL query: 'SELECT quarter_number,'.

Test Cases
Run SQL

Query 9

Query 10

Query 11

Query 12

Query 13

Query:

```

SELECT
  quarter_number,
  ROUND(AVG(julianday(ship_date) - julianday(order_date)),2) AS avg_days_to_ship
FROM order_t
GROUP BY quarter_number
ORDER BY quarter_number

```

Output:

Showing 4 rows

quarter_number	avg_days_to_ship
1	57.17
2	71.11
3	117.76
4	174.1

Observations and Insights:

- Average shipping time rises each quarter, growing from 57 days in Q1 to 174 days in Q4, indicating worsening fulfillment speed.
- Shipping time increases by over 116 days from Q1 to Q4, meaning customers wait nearly three times longer for delivery by the final quarter.
- While delays increase moderately from Q1 to Q2, they accelerate sharply in Q3 and Q4, suggesting operational strain later in the year.
- The increase in delivery time aligns with declining customer ratings and rising negative feedback, indicating that fulfillment delays may be impacting customer experience.

Business Metrics Overview

Total Revenue	Total Orders	Total Customers	Average Rating
Not counting discounts (total/gross revenue): 125482804.430	1000	994	3.135
Considering discounts (net revenue): 124714086.324			
Last Quarter Revenue	Last quarter Orders	Average Days to Ship	% Good Feedback
23346779.631	199	97.96	21.5

Note: These values must be derived using SQL queries. Some of them may have already been obtained while answering previous questions. (wasn't sure what exactly you guys considered 'total revenue' so I included both, I hope that is okay)

Business Recommendations

- Improve shipping and fulfillment operations to reduce delivery delays.
 - Average shipping time increased significantly from approximately 57 days in Q1 to approximately 174 days in Q4, indicating major operational slowdowns. The company should evaluate logistics partners they work with, optimize delivery routes, and consider different warehousing or domestic hubs to shorten the time it takes in transit. Implementing real time shipment tracking and delay notifications could also improve transparency to the customers if that is not in place already. Reducing fulfillment time is very likely to improve customer satisfaction, lower negative feedback, and support stronger repeat purchases.
- Strengthen after sales service and customer support experience.
 - Customer ratings declined steadily across all quarters, with Q4 satisfaction levels falling between 'Bad' and 'Okay', while negative feedback rose sharply. New-Wheels should expand customer support channels like live chat, call support, service ticketing, and maybe introduce post-delivery follow ups. Offering limited warranties or service guarantees on pre-owned vehicles could also build trust. After sales trust and strong relationships can greatly improve a company if done right.
- Launch customer retention and loyalty initiatives to stabilize demand.
 - Order volume and revenue declined each quarter, suggesting reduced repeat purchases or weakening customer engagement. The company should implement things such as loyalty programs, referral incentives, repeat buyer discounts, and even service credits for returning customers. Personalized email/app campaigns targeting past buyers could further bring in re-



engagement. Retaining existing customers is known to be more cost effective than getting new ones and can help stabilize sales performance during downturns. This could be introduced first before targeting new customers to ensure loyal ones are assessed as a priority.

- Introduce delivery experience tiers such as expedited shipping or premium fulfillment options. This is a bit more out of the box but could boost revenue in a new way and improve delivery times especially with customers who have had bad experiences.
 - Given the sharp rise in shipping delays, New-Wheels could implement tiered delivery options such as expedited shipping, premium handling, or guaranteed delivery windows for an added fee. Customers willing to pay for faster delivery could offset costs while improving satisfaction for time sensitive buyers. This approach not only could enhance customer choice but also creates an additional revenue stream while other improvements are underway.